

Problem Statement: Configurable Multimodal Semantic Validation System - Doc Vs Doc [GenAI; Multimodal; NLP; Semantic Reasoning]

1. Problem Background

Organizations across sectors (Healthcare, Finance, Legal, Logistics) face a bottleneck in validating information between heterogeneous document sets—such as comparing Health Insurance Benefit Grids against Summary of Benefits and Coverage (SBC), or Purchase Orders against Delivery Notes.

Traditional systems fall short because they rely on exact string matching or rigid rule-based templates. Real-world documents are often:

- **Multimodal in Nature:** Containing information locked in tables, scanned images, and complex layouts.
- **Semantically Diverse:** Using different terminology for the same concept (e.g., "Net 30" vs. "Payment due in one month").
- **Structurally Inconsistent:** Information may be located in different sections across document types.

2. Proposed Solution

The project aims to develop a **Generalized, Config-Driven Document-to-Document Validation System** powered by Multimodal Large Language Models (LLMs). Unlike "black-box" comparison tools, this system introduces a **User-Defined Entity Configuration Layer**, allowing domain experts to define exactly what needs to be validated, the semantic description of those entities, and optional few-shot examples for high-precision extraction.

The system will be **domain-agnostic**, achieving high performance in specialized fields (like medical or legal) not through model fine-tuning, but through **Contextual Prompt Engineering and Configuration-Based Grounding**.

3. Scope & Technical Workflow

The system will process both **E-printed and Scanned PDFs** through a unified multimodal pipeline:

1. **Ingestion & Perception:** Using Multimodal LLMs to perform **layout-aware parsing**. This eliminates the need for separate OCR engines and preserves the relationship between visual elements (tables, headers, signatures).
2. **Config-Driven Extraction:** The system reads a user-provided JSON/YAML configuration file containing:
 - **Entity Name:** (e.g., **Payment_Terms**)
 - **Entity Description:** (e.g., "The duration within which the invoice must be settled.")
3. **Semantic Reasoning Engine:** Instead of simple distance metrics, the system employs **Chain-of-Thought (CoT) reasoning** to compare extracted values. It evaluates if "Net 30" and "Due in 30 days" represent the same underlying business logic.
4. **Validation Output:** For every field, the system generates:
 - **Audit Status:** **Match**, **Mismatch**, **Partial Match**, or **Ineligible** (if the field is missing).
 - **Reasoning:** A natural language explanation of the AI's decision.
 - **Confidence Score (0-1):** A metric reflecting the model's certainty.

4. Reliability, Trust, and Hallucination Control

To ensure enterprise-grade reliability in high-stakes environments:

- **Evidence Grounding:** The system will be required to provide "citations" or coordinates from the source documents for every extraction, allowing human-in-the-loop verification.
- **Strict Instruction Set:** The LLM will operate under a "Restricted Output Schema," where it is penalized for making assumptions outside the provided document context.
- **Confidence Thresholding:** Any validation with a confidence score below a pre-set threshold (e.g., 0.85) will be flagged for manual review, effectively managing false positives.

5. Differentiation (Unique Value Proposition)

Most existing tools are either "Point Solutions" (only for invoices) or "Simple Diff Tools" (highlighting text changes).

Our solution differentiates itself by:

- **User Control:** Providing a low-code configuration file that allows a non-technical user to adapt the tool from a "Legal Contract Reviewer" to a "Medical Claim Validator" in minutes.
- **Semantic Intelligence:** Moving beyond "Search & Find" to "Understand & Validate," handling complex cross-lingual and cross-terminology discrepancies that traditional OCR/NLP pipelines miss.
- **Multimodal Native:** Handling scanned images and complex table structures as visual inputs rather than converting them to messy raw text.

5. Expected Outcomes

- A **functional end-to-end pipeline** that takes two heterogeneous documents and a configuration file as input.
- An **automated audit report** providing clear "Match" or "Mismatch" statuses with human-readable explanations.
- A significant **reduction in manual document verification time** for organizations in high-compliance sectors.